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TOPICAL REVIEW

PP-PQB: Privacy-Preserving in Post-Quantum Blockchain-Based Systems: A Systematization of Knowledge

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ABSTRACT Blockchain technology has produced effective solutions and provides security by using cryptographic tools for various applications, attracting attention from the academic community. Therefore, researchers have taken advantage of the features of blockchain technology to increase the security of the ecosystem. Recently, as the existence of quantum computers has been felt, researchers have started to benefit from post-quantum cryptography to increase privacy and security. There has been an increase in data and asset protection in post-quantum blockchain-based solutions. To the best of our knowledge, there is no comprehensive review or taxonomy that provides a complete picture of post-quantum secure structures with privacy-preserving techniques that have the potential to be used in blockchain. This paper aims to close this gap by systematically examining these approaches and revealing the deficiencies in the existing literature and the development potential in these areas. The taxonomy examines the role of blockchain technology in post-quantum cryptography and emphasizes the potential of technologies such as zero-knowledge proof to ensure privacy in post-quantum blockchain-based systems. We also review the existing literature on addressing the performance overhead, interoperability, scalability, and security challenges in implementing post-quantum cryptography in zero-knowledge proof-enabled blockchain architectures that protect against quantum computing threats. The studies are collected from journal papers in widely used academic databases between 2018 and 2024. The studies are subjected to certain elimination criteria, and 13 studies are reviewed in detail. Our approach will facilitate discussions on future research directions by proposing the accessibility of post-quantum cryptography against quantum threats to blockchain systems and solutions to the challenges that arise in the integration phase.

INDEX TERMS Blockchain, post-quantum cryptography, post-quantum blockchain, zero-knowledge proof.

I. INTRODUCTION

The development of blockchain technology, using cryptographic protocols, plays an important role in ensuring security and protecting confidentiality. Cryptography is a way of converting text into scrambled information, that is, into encrypted text, and whoever has the relevant key (private key) can decrypt the encrypted message. Among these, digital signatures are among the basic cryptographic primitives used to strengthen blockchain, providing features such as user privacy, account management, and consensus efficiency. Blockchain is a distributed shared database that, unlike a traditional database, stores hashes of data in cryptographically interconnected blocks. Its decentralization means that no person or group has the authority to change it, and the data is immutable. This structure ensures reliability in an untrusted environment with the immutability and non-repudiation of transactions. It has been proposed as

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an important technology in many areas, such as healthcare [1], supply chain [2], smart systems [3], e-voting [4], and digital identity verification [5]. Moreover, it provides various security mechanisms, including data protection and authentication for IoT applications [6]. In the past decade, various privacy-preserving blockchain models have been introduced to provide security. However, newly developed privacy analysis methods have reached a level that threatens the security of the systems [7], [8]. Moreover, recent studies have proposed innovative privacy-preserving mechanisms using blockchain technology, such as blockchain integration for privacy-preserving electric vehicle overhead prediction with federated learning [9] and scalability for IoT systems and solutions focusing on privacy-preserving mechanisms in blockchain-based federated learning systems [10]. Although this technology offers increasing advantages to address security concerns, it is under serious threat with the emergence of quantum computing technologies because existing blockchain platforms are based on traditional cryptosystems such as Rivest-Shamir-Adleman (RSA) [11] and the Elliptic Curve Digital Signature Algorithm (ECDSA) [12]. Advances in quantum computing make the resilience of traditional cryptographic protocols against quantum attacks questionable and pose a significant threat to the security of blockchain systems. In addition, while the process regarding the existence of Quantum computers remains uncertain, recent studies conducted by Google [13] and IBM [14] strengthen the opinion that post-quantum secure blockchain systems are needed.

Post-quantum cryptography (PQC) is a cryptography based on the assumption of a quantum computer [15]. The field of research is fairly new, and there has been a recent increase in work focusing on leveraging PQC to enhance privacy and the need for quantum resilience in decentralized applications. The main challenge here is to ensure cryptographic security and usability. Cryptographic methods such as lattice-based, code-based, multivariate, and isogeny-based play a role in preparing the blockchain for the quantum age, and post-quantum blockchain architectures are designed. Public-key cryptographic protocols such as RSA and Diffie-Hellman are based on the assumption that factoring a large integer, finding a sequence, and finding a discrete logarithm is difficult. Quantum computers are known to be able to solve any of these problems in polynomial time ($O(n^2)$ complexity) [16]. Quantum computers have the potential to break currently used schemes (such as elliptic curve cryptography (ECC)) with algorithms such as Shor's [17] algorithm. The underlying features of PQC that make it challenging include the following;

1) It is quantum resistant to quantum attacks with a different approach than traditional algorithms.
2) Instead of relying on the challenge of solving the discrete logarithm problem, it has complex mathematical structures such as multivariate polynomials, mixed basis structures, lattices, etc., and it relies on the challenging mathematical problem.

The National Institute of Standards and Technology (NIST) launched a standardization competition in 2016 to determine cryptographic algorithms that will remain secure against quantum computers. This competition, conducted in three stages, reached its final phase between 2022 and 2024, leading to the announcement of the following standards: CRYSTALS-Kyber [18], Dilithium [19], FALCON [20], and SPHINCS+ [21]. These algorithms are crucial for mitigating potential threats posed by quantum computing. NIST's PQC Project [22] has concluded, with CRYSTALS-Kyber selected as the post-quantum Key Encapsulation Mechanism (KEM) and Dilithium, Falcon, and SPHINCS+ chosen as post-quantum digital signature algorithms. Dilithium and Falcon have also been tested by Hardware Security Module (HSM) vendors to verify digital signature creation, key import during authentication, and signature verification. While standardization efforts continue, it has been concluded that, for certain use cases, digesting the message before passing it to the signing algorithm, or using pure PQ signatures without digests, may be preferable.

The fact that blockchain technology has many application areas indicates that there is a need for PQ secure structures due to the existence of known quantum computers. Furthermore, the developments in integrating Big Data and Artificial Intelligence (AI) with blockchain systems [23], such as increasing data processing, scalability, and security, play an important role in meeting the increasing demands of PP-PQB solutions. To the best of our knowledge, there is no Systematization of Information (SoK) that provides a complete picture of PQ secure structures with privacy-preserving techniques that have the potential to be used in blockchain. Therefore, this paper, the first we know of, fills this gap by reviewing approaches used for privacy-preserving PQB (PP-PQB) applications required by blockchain applications. It also examines the existing literature on the current challenges and proposes solutions.

This section addresses the need for integrating blockchain technology with PQC. It discusses the vulnerability of traditional cryptographic systems to quantum threats and the potential for enhancing the security of PQB systems. In addition, the aim of this study is to fill the gaps in the literature.

A. RESEARCH MOTIVATION

In recent years, since quantum computing threatens the security of applications in traditional cryptography, blockchain technology has emerged as a promising solution to ensure the security of transactions. However, even if traditional blockchain systems are robust, the existence of quantum computers, especially due to their computational power exceeding classical computational methods, threatens their security against quantum attacks. Since the basis of blockchain technology is the existing classical cryptographic protocols, quantum computers have a high potential to weaken the

security of blockchain networks. Therefore, the need for post-quantum blockchain applications arises from the following reasons.

1) In blockchain systems, classical cryptographic protocols such as asymmetric encryption and hash functions (RSA, SHA-256) pose a cryptographic security threat because they can be easily broken by Shor's and Grover's algorithms [24].
2) The ability to solve authentication schemes to prevent unauthorized access and fraudulent transactions with quantum computers [25].
3) Traditional blockchain systems, which block user identities and meet the need for anonymity and privacy, can be solved in the presence of quantum computers [26].
4) Quantum computers negatively affect the permanent data security and reliability of blockchain systems used for sustainable systems [27].

PQB applications and the provision of structures such as security, anonymity, privacy, and integrity are necessary to ensure the sustainability of this technology. With the help of PQC, new methods for digital signature schemes and KEMs have appeared, relying on cryptographic structures based on hard-to-solve mathematical problems, such as lattice-based, hash-based, and code-based approaches. Due to these main changes, post-quantum security can be provided by applying them to blockchain applications. Therefore, there is a need to develop blockchain architectures that aim to provide quantum-resistant security and privacy against quantum threats by taking advantage of advances privacy-preserving techniques in PQC algorithms such as lattice-based cryptography, code-based cryptography, and hash-based cryptography. When looking at the PQB applications in the literature, although there is no full PQB application, some studies have taken important steps in this direction [28]. In [29], the authors designed a PQB system for smart city applications. The study developed a post-quantum Proof-of-Work (PoW) consensus algorithm using multivariate quadratic equations and an identity-based signature scheme for a lightweight and fast transaction structure. However, it has been observed that the blockchain transaction security and PoW consensus protocol components of the system have developed post-quantum, and the basic structure of the system, such as data structures and transaction verification process, have not been fully post-quantum. In [30], the authors proposed a post-quantum ring signature-based blockchain architecture for the Internet of Social Things. The proposed management emphasized its resistance to classical and quantum computers. In the study, anonymity and privacy were provided by using ring signatures, and security was provided by using ring signatures and multivariate quadratic equations. The study designed certain system components to provide post-quantum security and observed that a complete PQB structure was not provided. In [31], the authors proposed a lattice-based signature scheme to secure the blockchain network. In the study, private and public keys were generated using Bonsai Trees technology. The current study also has designed certain components to provide post-quantum security.

Therefore, while PQB-focused studies in existing surveys focus on security, privacy-preserving methods (such as ZKP) in PQB systems have not been thoroughly examined. Furthermore, the performance of these methods and the privacy-performance trade-off have not been discussed in depth.

B. RESEARCH CONTRIBUTION

Various surveys have been conducted on post-quantum secure blockchain structures [26], [32], [33], but there is a lack of systematic analysis on privacy-preserving PQC-integrated blockchain. In this study, we present a survey about PP-PQB systems. This study, compared to other studies in the literature, not only remains in the theoretical framework but also fills the gap in the literature by analyzing how PP-PQB solutions can be used in real-world applications. We propose the optimization techniques to reduce the computational load without compromising security for the balance between performance and security, which is often overlooked in the existing literature. We also provide a research contribution analyzing the effects of ZKP on scalability and performance in post-quantum systems, which is felt lacking in this field.

1) This is the first study focusing on the performance overhead, interoperability, scalability, and security (OISS) challenges of PP-PQB solutions.
2) We examine PP-PQB systems to fill the gap in the literature in this area and analyze the role and performance of privacy-preserving techniques in providing privacy.
3) We analyze the usage cases of PP-PQB applications and evaluate their efficiency.
4) We discuss the security constraints of the proposals and the important application areas of the architectures.
5) We present a review of applications that include ZKP, which ensures the confidentiality and integrity of transactions without compromising privacy in post-quantum blockchain (PQB) applications.
6) We provide suggestions on the practical applicability of post-quantum privacy-preserving protocols, especially theoretical studies.
7) We propose new approaches to improve the security and efficiency of PQC-integrated Blockchain architectures.
8) We give new research directions on the development and improvement of privacy-preserving methods in PQB, which are lacking in the literature.
9) We present a roadmap on quantum-resistant Layer2 solutions and integration of Multi-chain models to solve scalability and efficiency problems.

C. ORGANIZATION

This review addresses the critical gap in blockchain systems by investigating the integration of quantum-resistant cryptographic primitives and privacy-preserving techniques (PP-PQB) to enhance security against quantum attacks.

We systematically collected and analyzed studies from major academic databases, including Scopus, WoS, IEEE Xplore, ScienceDirect, Springer Link, and ACM Digital Library, and applied selection and elimination criteria to ensure relevance to our research questions. The review provides an SoK as follows:

In Section II, we discuss the systematic process for the studies to be reviewed and summarize those selected. In section III, we explain the basics of the PQC and blockchain concept and the perspectives of the elements that provide security. Then, we present the use of privacy-preserving techniques (such as ZKP) in PQC approaches and the modules that need to be changed in the blockchain for PQB. In Section IV, we briefly review proposed PP-PQB solutions from the systematic process and solutions to several identified challenges to improve security against quantum attacks. In Section V, we discuss the vulnerability of selected reviews to quantum attacks of cryptographic primitives used in blockchain systems, present threats to validity, and future research directions. Finally, we present the conclusion.

D. RELATED LITERATURE

This section examines post-quantum blockchain-based review studies published between 2018-2024. We obtain the review studies through specific filtering, which we present in detail in the following sections. Although various surveys and studies have been conducted on PQB [32], [33], [34], it has been observed that PP-PQB studies are quite new, and the applicability and comprehensive discussion of privacy-preserving techniques are limited. Focusing on privacy-preserving methods in PQB systems, the existing literature is reviewed, and the gaps in this field for post-quantum secure blockchain systems are addressed. Table 1 presents existing reviews of PQB studies and their main differences. We show different emphases of studies and do not indicate superiority or inferiority.

Fernandez-Carames and Fraga-Lamas [32] examined how PQC can be applied to blockchain applications, focusing mainly on encryption and traditional signature schemes. They analyzed the PQC systems most suitable for blockchain and the challenges of these systems. In the study, the authors investigated various technical aspects related to the vulnerability of blockchain technology to quantum attacks. To briefly state these, they examined how algorithms such as Grover's and Shor's used by quantum computers pose threats to traditional public-key cryptography and hash functions such as RSA, ECDSA, and SHA-256 used in blockchain. They also examined lattice-based, code-based, multivariate, and hash-based post-quantum cryptography methods designed to resist quantum attacks. They presented comparisons of the performance, key sizes, and security levels of post-quantum public-key cryptography and digital signature schemes such as McEliece, NewHope and It also focuses on the challenges posed by the key size, key generation time, and computational resource demands of integrating PQC into blockchain systems.

TABLE 1. Comparison to Review Studies.

	Year	PQC	PQB	ZKP	PQZKP	PP-PQB	Systematic
[32]	2020	Yes	Yes	No	No	No	No
[34]	2023	Yes	Yes	No	No	No	Yes
[33]	2022	Yes	Yes	Yes	No	No	Yes
[35]	2023	Yes	Yes	Yes	No	No	No
[27]	2023	Yes	Yes	No	No	No	No
[36]	2024	Yes	Yes	No	No	No	No
Ours	-	Yes	Yes	Yes	Yes	Yes	Yes

Gomes et al. [34] used a systematic method to analyze PQ solutions' effects on blockchain consensus. They examined the effects of improvements in security, scalability, trust, and privacy on PQB consensus. In the study, the authors examined how existing consensus protocols threaten quantum attacks, detailed post-quantum blockchain consensus solutions, and investigated PQC-based solutions such as quantum random number generation, quantum entanglement, quantum key distribution, and multivariate polynomial equations. They discussed the development of consensus algorithms based on quantum physical processes using random number generation methods that ensure that random numbers in the blockchain are verifiable and reproducible. They examined how quantum entanglement and quantum measurement methods, which transfer and verify data over quantum secure channels, can improve blockchain security and verification processes.

Buser et al. [33] reviewed existing research in the literature on post-quantum exotic signature schemes rather than discussing specific post-quantum ordinary signature proposals used in blockchain systems. They also examined schemes that can be used to increase security and privacy for blockchain systems. They also reviewed schemes that can increase security and privacy for blockchain systems. In the study, the authors specifically considered advanced signature types: multi-aggregate, threshold, adaptor, blind, and ring signatures. They evaluated how account management and consensus efficiency can be used to increase privacy in blockchain systems without smart contract support. They detailed the security, efficiency, and impact of post-quantum signature algorithms on blockchains and indicated shortcomings and future research directions.

Yang et al. [35] conducted comprehensive general literature research on post-quantum and quantum blockchains and provided a comparative study. They also discussed the cryptographic foundations used in post-quantum blockchains against quantum attacks and investigated their vulnerabilities and challenges. In the study, the authors examined how systems using classical cryptographic protocols should be strengthened against quantum computer attacks and analyzed lattice, code-based, isogeny-based post-quantum cryptographic protocols. They discussed how hybrid quantum blockchains can be built based on quantum technologies such as quantum key distribution, quantum signatures, and quantum teleportation. They examined and analyzed the

TABLE 2. Research question list.

	Research Questions
RQ1	What are the main methods for PP-PQB?
RQ2	Which methods are used to improve confidentiality and integrity in PP-PQB applications?
RQ3	What are the effects of the methods applied to PP-PQB applications on OISS?
RQ4	What are the future challenges in adopting and implementing privacy-preserving structures, including ZKP, in PQB applications?

studies on how existing blockchain systems can be protected against quantum threats by focusing on PQB.

Thanalakshmi et al. [27] examined the inclusion of post-quantum signature schemes in the blockchain system into the "Interplanetary File System (IPFS)" integrated with the blockchain system. They also discussed the comparison of post-quantum signature schemes proposed by the "National Institute of Standards and Technology (NIST)" with the "Elliptic Curve Digital Signature Algorithm (ECDSA)". In the study, the authors examined the resistance of the Dilithium, FALCON, and SPHINCS+ signature schemes proposed by NIST against quantum attacks and their performance, such as key generation time, signing time, verification time, and key/signature sizes. They also investigated how IPFS can be used to reduce the overhead of these signature schemes with large key and signature sizes in the blockchain.

Gharavi et al. [36] examined PQB types in blockchain-based IoT applications and discussed the current standard foundations of PQB studies and their applicability in IoT. In the study, the authors detailed how classical cryptographic protocols such as RSA and ECDSA can be broken by quantum computers and how the resistance of existing systems against quantum attacks can be changed. They analyzed the integration and performance of various post-quantum cryptographic methods, such as hash-based, lattice-based, multivariate-based, and isogeny-based cryptographic systems into IoT devices. They also investigated the integration of PQB systems into IoT systems and examined their effects on security and performance. Specifically, the authors emphasized that the lattice-based post-quantum cryptographic method is more efficient in IoT devices.

This section presents a literature review of existing PP-PQB techniques. The general trends, strengths, and weaknesses of the studies are identified, and research gaps in these areas are highlighted.

II. RESEARCH METHODOLOGY

This section includes selecting and filtering studies on a PP-PQB between 2018 and 2024. Research questions (RQ) are determined according to the research objectives. In detail, it includes the RQ and the topics covered in the research field, as well as their selection and elimination by filtering from various academic databases. The RQ of the study is presented in Table 2.

To identify research studies related to PP-PQB, Web of Science (WoS), Scopus, IEEE Xplore, Springer Link, ScienceDirect, and ACM Digital Library databases are scanned. The search array was run on each scientific repository to execute the query expressions. Advanced search tools of Scopus, WoS, IEEE Xplore, and ACM Digital Library databases are used. The main search engine uses relevant query expressions in Springer Link and ScienceDirect databases. Query expressions used in databases are shown in Table 3. It needs to be filtered to determine which study to review. In filtering, certain selection and exclusion criteria (EC) are defined. Firstly, we exclude PQB and all cheat sheets after searching PP-PQB. To exclude the remainder, we use other EC (Table 4). The first stage is about reading the title and abstract of the papers. Then, the introduction and conclusion parts of the papers are read, and finally, it is read in its entirety and analyzed according to the research questions.

In the first step, 982 journal papers are obtained in the field of PQB, including 92 articles in Scopus, 34 in WoS, 43 in IEEE Xplore, 29 in ACM Digital Library, 301 in Springer Link, and 483 in ScienceDirect. 477 journal papers related to PP-PQB are obtained, including 18 journal papers in WOS, 18 in Scopus, 14 in IEEE Xplore, 46 in ACM Digital Library, 134 in Springer Link, and 247 in ScienceDirect. The studies are evaluated according to their suitability for post-quantum cryptographic methods, (blockchain-based) privacy-preserving mechanisms, quantum resistance, blockchain performance, and scalability. In the first screening, 505 journal papers are excluded from papers lacking technical depth, studies focusing only on classical cryptography, and papers published outside of English.

The remaining 459 journal papers are excluded through title and abstract reading analysis, and duplicate journal papers are excluded in the subsequent elimination. These studies do not directly answer our RQ or lack experimental validation.

In the next stage, the introduction and conclusion are read in the selected journal papers, and non-relevant journal papers are excluded. As a result of all these processes, 13 journal papers are reviewed in detail within the scope of Q1 and Q2 journals, which present innovations focusing on privacy and security in PQB systems. The contribution level of the studies is analyzed based on factors such as case studies and experimental results. In addition, some of the selected studies offer solutions to ZKP, lattice-based cryptography, and scalability problems. Figure 1 systematically shows the filtering stages of the study.

III. POST-QUANTUM BLOCKCHAIN

A. POST-QUANTUM CRYPTOGRAPHY

PQC has been developed with quantum computers to protect against attackers. PQC is also a classical cryptography based on NP-hard mathematical problems [37]. Symmetric

TABLE 3. Query statements.

Database	Statement
WOS	TI=(privacy-preserving post-quantum blockchain) OR TI=(post quantum zero knowledge proof blockchain) OR TI=(post-quantum zero knowledge proof blockchain) OR TI=(post-quantum zkp blockchain) OR TI=(post quantum zkp blockchain) OR TI=(pqzkp blockchain) OR AB=(privacy-preserving post-quantum blockchain) OR AB=(post-quantum zkp blockchain) OR AB=(post-quantum zero knowledge proof blockchain) OR AB=(post quantum zero knowledge proof blockchain) OR AB=(pqzkp blockchain) OR AK=(privacy-preserving post-quantum blockchain) OR AK=(post-quantum zero knowledge proof blockchain) OR AK=(post quantum zero knowledge proof blockchain) OR AK=(post quantum zkp blockchain) OR AK=(post-quantum zkp blockchain) OR AK=(pqzkp blockchain)
Scopus	TITLE-ABS-KEY (privacy-preserving AND post-quantum AND blockchain OR post-quantum AND zero-knowledge AND proof AND blockchain OR post AND quantum AND zero-knowledge AND proof AND blockchain OR post-quantum AND zero AND knowledge AND proof AND blockchain OR post-quantum AND zkp AND blockchain OR post AND quantum AND zkp AND blockchain OR pqzkp AND blockhain)
IEEE Xplore	("All Metadata": privacy-preserving post-quantum blockchain) OR ("All Metadata":post-quantum zero knowledge proof blockchain) OR ("All Metadata":post quantum zero knowledge proof blockchain) OR ("All Metadata":post-quantum zkp blockchain) OR ("All Metadata":post quantum zkp blockchain) OR ("All Metadata":pqzkp blockchain)
ScienceDirect	“privacy-preserving post-quantum blockchain” OR "post-quantum zero knowledge proof blockchain" OR "post quantum zero knowledge proof blockchain" OR "post-quantum zkp blockchain" OR "post quantum zkp blockchain" OR “pqzkp blockchain”
ACM Digital Library	Title:(“privacy-preserving post-quantum blockchain” OR "post-quantum zero knowledge proof blockchain" OR "post quantum zero knowledge proof blockchain" OR "post-quantum zkp blockchain" OR "post quantum zkp blockchain" OR “pqzkp blockchain”) OR Abstract:(“privacy-preserving post-quantum blockchain” OR "post-quantum zero knowledge proof blockchain" OR "post quantum zero knowledge proof blockchain" OR "post-quantum zkp blockchain" OR "post quantum zkp blockchain" OR “pqzkp blockchain”) OR Keyword: (“privacy-preserving post-quantum blockchain” OR "post-quantum zero knowledge proof blockchain" OR "post quantum zero knowledge proof blockchain" OR "post-quantum zkp blockchain" OR "post quantum zkp blockchain" OR “pqzkp blockchain”)
Springer Link	(“privacy-preserving post-quantum blockchain” OR "post-quantum zero knowledge proof blockchain" OR "post quantum zero knowledge proof blockchain" OR "post-quantum zkp blockchain" OR "post quantum zkp blockchain" OR “pqzkp blockchain”)

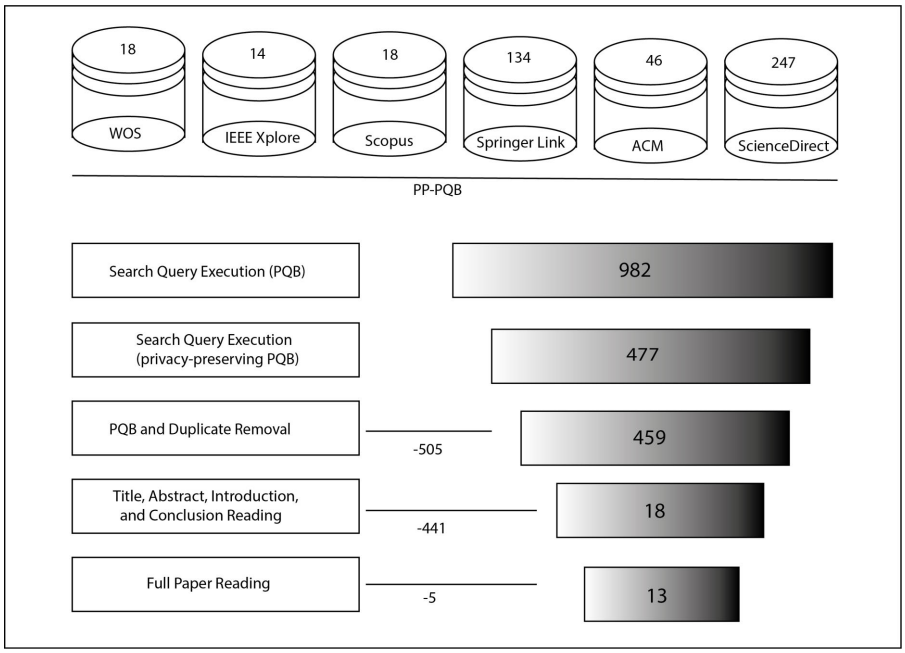

FIGURE 1. Systematic review/filtering process.

Algorithms and hash functions provide relative security for post-quantum. However, researchers in this field have to look for new number theory problems that will base the security of asymmetric algorithms, that is, new mathematical

TABLE 4. Exclusion criteria.

	Exclusion criteria
EC1	Papers whose main topic is not specifically PP-PQB
EC2	Papers that do not provide solutions for privacy-preserving implementations in PQB
EC3	Papers which are not Journal papers
EC4	Papers not written in English

problems that quantum computers cannot easily solve. Shor's algorithm [17] is proposed to solve the discrete logarithm problem on a quantum computer by executing parallel operations on the processor. On the other hand, Grover's algorithm [38] can speed up search operations according to the complexity of attacks $\sqrt{n}$ (n is the size of the brute force search domain). This situation threatens the security of the symmetric and hash functions.

The National Institute of Standards and Technology (NIST) announced the PQC standard algorithms in 2022, continuing a process it started in 2016 to prevent quantum attacks. In the literature, different PQC approaches such as lattice-based, code-based, isogeny-based, hash-based and multivariate-based are discussed for data privacy and security for post-quantum systems. Lattice-based systems, which are especially prominent in signing and key encapsulation, and hash-based systems used in signing are used for privacy-preserving in post-quantum secure blockchain systems.

Lattice-based algorithms were firstly proposed by Ajtai [39] for the development of reliable cryptosystems based on the NP-hard [40] mathematical lattice problem. While other cryptosystems include security based on average-case problems, lattice-based algorithms have the feature of providing security based on worst-case problems. Lattice-based algorithms are based on the nearest (CVP) or shortest vector (SVP) problem, which is a special arrangement of points with a periodic structure in the n-dimensional vector space. The cryptographic generators used in these cryptographic systems provide security proof based on worst-case hardness and are highly efficient [41]. Therefore, lattice-based cryptography is considered highly promising for post-quantum encryption. A special category of hard problems, known as Learning With Errors (LWE), also forms the basis of various lattice-based cryptosystems, such as R-LWE and LP-LWE. In response to the NIST call, schemes like Kyber [42], FALCON [20], and Dilithium [19] were accepted as standards for key encapsulation and digital signatures. In PQB applications, the ZKP structure can be used on Short Integer Solution (SIS) and LWE problems when using lattice-based structures, and Merkle Signature Schemes (MSS) and their derivatives when using hash-based structures. Multivariate-based structures can be used to protect the privacy of the signer, and isogeny-based structures can be used in structures that require verification in secret information sharing.

The hash-based digital signature scheme is a type of post-quantum encryption scheme used as an alternative to today's digital signature schemes. Its security is based on certain mathematical problems that provide collision resistance and preimage resistance, which is different from other cryptosystems. Finding collisions and preimages is a challenging problem due to the underlying hash function. Therefore, it is a signature scheme that can be used in authentication. It takes advantage of the hash function, which is widely used in cryptosystems, and protects against quantum attacks due to its unidirectionality, preimage, and collision resistance properties. The SPHINCS+ and Gravity-SPHINCS schemes selected by NIST are based on a stateless approach. Among the stateful class, there are also variants such as Witernitz [43], Lamport's given One-Time Signature (OTS) scheme, WOTS+ [44], and WOTSPRF [45]. Although they attract attention with their structures resistant to attacks by quantum computers, they have some limitations regarding scalability, efficiency, and application flexibility. A larger signature size than other PQC algorithms increases storage and communication costs in IoT networks with limited resources. Since these signature schemes require stateful, a key must be used in each signature. Therefore, if the key is reused, the security of the system is at risk. In addition, due to its structure, intensive operations in creating and verifying each signature complicate the key management process and may have difficulty supporting high transaction volumes in large-scale blockchain applications. These lattice-based and hash-based algorithms, which are discussed in the context of PQC, offer different security and performance characteristics [46]. When choosing between these approaches, key size, signature size, transaction time, and energy efficiency should be considered. However, even if the use of ZKP in symmetric and code-based structures is theoretically possible, it is not possible to design and implement due to limitations such as key management challenges, ZKP complexity, key size, and performance cost. Therefore, the use of ZKP in symmetric and code-based structures is limited in the current literature, and there are potential research opportunities.

This section examines hash-based and lattice-based cryptographic algorithms, comparing their performance metrics and application areas. It details the strong security features of hash-based algorithms and their impact on large data sets, as well as the scalability and processing power advantages of lattice-based algorithms.

B. POST-QUANTUM BLOCKCHAIN FEATURES

Blockchain technology is a distributed ledger technology that provides immutability, transparency, and traceability of data and distributed peer-to-peer trust [47]. Consensus algorithms, distributed ledger technology, and decentralized structure are the basic components of the system. The security of blockchain technology is based on asymmetric cryptography and hash functions. Data security and immutability in the network are based on consensus protocols and classical encryption techniques (such as RSA [11] and ECDSA [48])

[49]. Figure 2 shows the generation of the blockchain and an example of a transaction in the generated chain. Since transactions are open and transparent in blockchain, ZKP technology ensures data confidentiality and proves transaction accuracy. This method protects privacy and secures the transaction verification process. It is an important technique that increases the security and efficiency of the system, especially in cases where private data needs to be protected and verified. Digital signatures used in blockchain systems are indispensable for identity verification, data integrity, and transaction non-repudiation. They verify the source of a transaction while ensuring that the data has not been altered. Various signature schemes, such as Multi-Signature [50], Blind Signature [51], Ring Signature [52], and Threshold Signature, are applied to blockchain platforms to meet the needs for security, anonymity, and scalability. These schemes increase transaction security in blockchain systems while providing higher levels of anonymity and privacy [53]. Algorithms such as RSA and ECC, which are widely used for digital signatures and key exchange, and hash functions such as SHA-256 and SHA-3 (Keccak), which form the basis of block hash operations and Proof of Work (PoW) mechanisms, are essential cryptographic tools in blockchain systems. Additionally, symmetric encryption algorithms like AES are used for data encryption in some blockchain applications, and digital signatures such as ECDSA, which provide authentication and non-repudiation, play a crucial role in enhancing blockchain security. In these tools, SHA-256 is the basic hash function used in many blockchains such as Bitcoin and performs many tasks such as transaction verification, creation of Merkle tree structures, and preservation of data integrity by hashing the data. SHA-256 provides data integrity and transaction verification processes by producing a fixed-length digest (hash) regardless of the input size. In addition, SHA-256 plays a critical role in transaction verification and chain security in Bitcoin's PoW mechanism, and this function ensures that a unique hash value represents each block in the blockchain chain.

The SHA-3 algorithm, selected as a standard by NIST in 2012, was designed to be secure the potential weaknesses of SHA-256. SHA-3 provides faster transactions and higher security by using sponge functions. It can be used in many blockchain applications as an alternative to SHA-256. ECDSA is a digital signature algorithm used to meet the authentication and non-repudiation requirements of blockchain transactions. It is widely used as a security layer, especially in popular blockchain platforms such as Bitcoin and Ethereum. Symmetric encryption algorithms such as AES ensure data confidentiality in blockchain applications. In particular, AES is preferred in IoT-based blockchain applications due to its low processing power requirement and high-speed advantage. Asymmetric encryption protocols such as Diffie-Hellman and RSA provide secure key exchange between users in the blockchain network. These protocols are critical for secure communication, especially for smart contracts and decentralized applications. However, quantum computers can quickly solve problems based on these algorithms thanks to Shor's and Grover's algorithms, and this threatens the security of cryptographic algorithms used in the blockchain infrastructure [54]. Therefore, the privacy-preserving techniques used in PQB both protect data integrity and provide a structure resistant to future quantum threats.

Real-world applications of PP-PQB systems highlight their critical role in addressing privacy and security challenges. In finance, ZKP-based protocols enable anonymous payment systems and ensure regulatory compliance without compromising user privacy. In particular, zk-SNARKs provide privacy and security for PP-PQB systems by allowing users to verify the validity of transactions without revealing account details. They are also used in financial applications such as anonymous payment systems. It can also be used in PP-PQB systems to provide regulatory authorities with the necessary information to prove the authenticity of only the required credentials in anti-money laundering processes [55]. Similarly, in healthcare, using ZKP protocols in PP-PQB systems enables the secure sharing of medical data while preserving privacy, allowing patients to verify their eligibility for treatments or participate in research without disclosing their full medical history [56]. This integration also addresses scalability and computational cost challenges while providing opportunities for innovation. These specific examples demonstrate the practical impact of integrating PP-PQB systems into existing infrastructures.

Definition 1: PQB is a distributed ledger structure designed to be resistant to attacks while preserving data integrity and confidentiality. It provides enhanced security against quantum attacks over pre-quantum methods such as RSA and ECDSA by integrating post-quantum cryptographic methods such as lattice-based and hash-based cryptography.

Definition 2: ZKP is a cryptographic technique built on three key properties: completeness, soundness, and zero-knowledge, which allow a party to prove the validity of a statement without revealing any additional information [53], [57]. In PQB systems, ZKP guarantees transaction privacy and data integrity while maintaining scalability and performance.

The main modules used in the blockchain, such as digital signature, key exchange protocols, hash functions, consensus mechanism, and smart contract protocols, need to be updated to resist quantum attacks [36], [58]. Instead of classical signature algorithms such as RSA and ECDSA used in digital signatures, PQ secure signature schemes such as NIST-approved Dilithium, FALCON, or SPHINCS+ should be used [59]. Instead of classical key exchange protocols such as Diffie-Hellman, PQ secure key exchange protocols such as Kyber or NTRU should be used [32]. In case of the possibility of breaking hash functions with Grover's algorithm, it is necessary to work with larger bit lengths in hash functions such as SHA-256 or SHA-3. However, the

integration of post-quantum blockchain needs to integrate with the following key challenges:

1) Signature and hash length [60]
2) Key size [60]
3) Computational complexity [61]
4) Execution and Energy consumption [62]

In PQB integration, features such as small signature and hash length, small key size, low computational complexity, fast execution, and low energy consumption are important challenges in PQC [63]. These challenges manifest in many areas, from standardization processes to efficient different-platform implementation. Since lattice-based algorithms require high processing power for intensive operations such as matrix multiplications and modular arithmetic (complex operations such as large key sizes, multiplication, and modular reduction of large polynomials used in encryption/decryption), it causes the block verification process to take a long time in large-scale blockchain networks and reduce the transaction throughput of the network, thus causing high computational costs. In addition, challenges such as the large key size requiring more bandwidth during transaction submission and increasing storage load make integrating these lattice-based structures into existing blockchain-based systems difficult. This causes compatibility problems and, in this case, increased transaction time or resource consumption and, thus, integration complexity. In general, it is also critical to establish international standards for the effective use of lattice-based algorithms. Choosing among PQC algorithms such as Kyber and Dilithium and adapting these algorithms to various use cases is a complex process. Blockchain networks can be deployed on different device types and hardware platforms, such as IoT devices. To support this diversity, lattice-based algorithms should be optimized. Small key pairs must be used to reduce complexity and reduce storage space for limited Internet of Things (IoT) end devices interacting in blockchain-integrated systems. In blockchain architecture, user signatures and data transactions in the data or block hash are stored. Therefore, the blockchain size increases as the signature and hash length increase. In such cases, a short signature and hash length should be used. For transactions to be carried out quickly in the blockchain, post-quantum cryptography must be as fast as possible. Therefore, choosing the parameters used correctly to ensure security and performance balance is a great challenge. Wrong parameter selection can lead to security vulnerabilities and negatively affect the transaction costs of the algorithm. The selected parameters are important for fast execution, to prevent resource-constrained devices from being excluded from blockchain-integrated transactions, and to provide low computational complexity and overhead. In addition, the number of transactions performed, the hardware used, and the security plans to be implemented affect power consumption. Fast execution provides low communication costs and overhead and impacts power consumption. This contributes to the efficient operation of the system. In addition, to enhance the privacy and security of blockchain ecosystems, post-quantum cryptographic algorithms such as Kyber and Dilithium, which provide resistance against quantum attacks by exploiting the difficulty of problems such as LWE and MLWE, will be the alternatives of traditional public-key cryptosystems such as ECDSA used in blockchain ecosystems. The signature schemes ensure data integrity and authentication processes in blockchain networks. Algorithms such as Dilithium and Falcon provide reliability in signing and verification processes and offer faster and more efficient architecture with low processing overhead. These features enable integrating IoT devices with blockchain systems, especially those with energy constraints [64]. However, implementing these algorithms at the protocol level creates technical complexity, especially in blockchain-based systems that require data privacy and transaction security. Protocols must be designed securely and scalable to achieve an optimal structure in transaction costs. The hash-based and lattice-based cryptographic schemes ensure data integrity and secrecy by resisting quantum attacks, such as Shor's algorithm for breaking RSA and ECC, and Grover's algorithm, which only weakens symmetric systems by effectively halving the key length. Additionally, using data blinding operations with lattice-based algorithms enhances user privacy and helps protect sensitive data. Such mechanisms support both decentralized authentication processes and secure data sharing [32]. Moreover, post-quantum cryptography enables secure consensus protocols by providing robust digital signatures and key exchange mechanisms in the blockchain ecosystem. Therefore, the overall privacy and integrity of the system against future threats posed by quantum computing is enhanced.

This section examines post-quantum security methods and privacy mechanisms used in blockchain technology. It also exemplifies the role of ZKP in system integration and the applications of these mechanisms in various sectors.

C. THE USE OF ZKP IN POST-QUANTUM SIGNATURES

ZKP is the protocol that allows one party (the prover) to convince the other party (the verifier) of the validity of the relevant statement without informing the other party (the verifier). It has the basic properties of completeness, robustness, and zero-knowledge [65]. These properties can be categorized into two main types based on interaction:

1) **Interactive:** Participating in an iterative protocol for the engagement of the statement by the prover and verifier, involving query-response requests through multiple rounds of interaction.
2) **Non-interactive:** the case where the prover does not require direct communication with the verifier to verify the statement.

The remaining properties are defined as follows:

1) **Completeness:** in the truth of the statement, the prover convinces the verifier.
2) **Robustness**: if the statement is untrue, the fraudulent prover cannot convince the verifier.

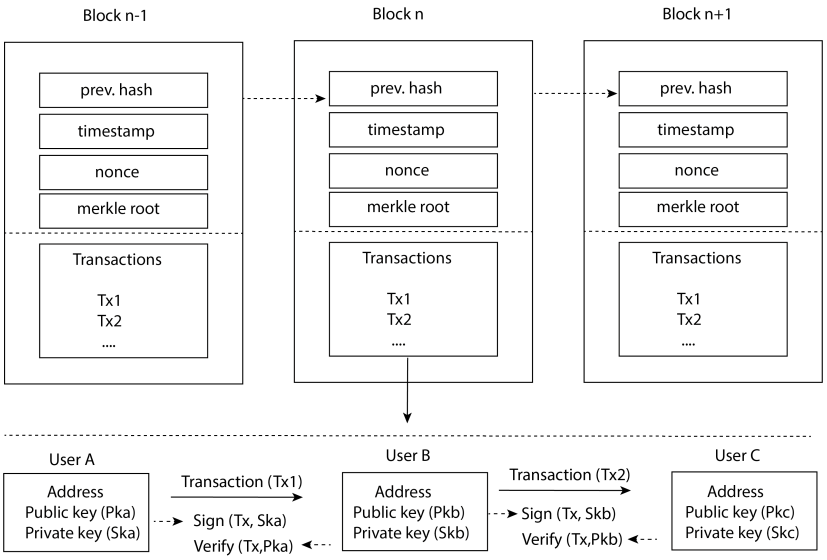

FIGURE 2. The sequence example of transactions in the blockchain (Simplified).

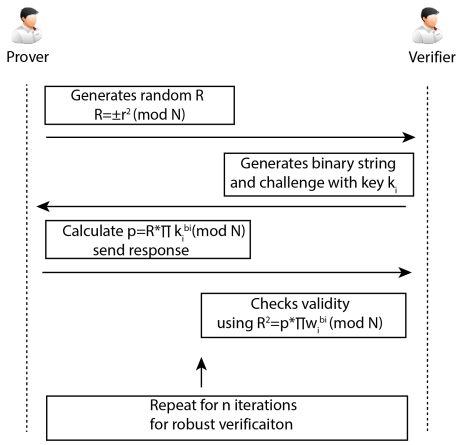

FIGURE 3. The working principle of Interactive-ZKP.

3) **Zero-knowledge:** when the statement is true, the verifier cannot learn any information other than the truth of the statement.

The interactive-ZKP is recalled in Figure 3 [65]:

1) Step 1: The prover generates a random R to pass $R = \pm r^2(modN)$ to the verifier.
2) Step 2: The verifier generates the binary string $(b_0, b_1, \ldots, b_{m-1})$ to generate the challenge and sends it to the prover along with the k_i key corresponding to the string.
3) Step 3: The prover calculates $\pi^2 y = p$ $(1 \leq y \leq n)$ using the $R \cdot \prod_{k_i}^{b_i}(modN)$ $(1 \leq i \leq m-1)$ formula and sends it to the verifier.
4) Step 4: The verifier checks the validity using the formula $R^2 = p \cdot \prod_{w_i}^{b_i}(modN)(1 \leq i \leq m-1)$, and if the equation holds, the iteration is considered.
5) Steps 1-4 are repeated for n iterations for robust verification.

There are non-interactive-ZKP protocols called Zero-Knowledge Short, Non-Interactive Information Argument (zk-SNARK), and Zero-Knowledge Scalable Transparent Information Arguments (zk-STARK), which have recently been researched for use in PQB applications and are widely used in Blockchain applications. These protocols offer potential capabilities for robust privacy and security. zk-SNARK utilizes ECC to generate keys and process transactions, providing proof of the validity of transactions while protecting user privacy. The working principle of zk-SNARK is shown in Figure 4.

1) Step 1: KeyGen (p_k, v_k), Proof (p), and verification (v) keys are created in the given circuit$C$.
2) Step 2: Proof $(p_k, x, a) \rightarrow \pi$, to generate evidence of the validity of$a$given statement $x$given p_k given a particular witness.
3) Step 3: Verify$(vk, x, \pi) \rightarrow b$, verifying the validity of the proof π on the given statement x. A boolean value indicating the validity of the proof b is 1 if successful, 0 otherwise.

zk-STARK is a variant of ZKP that addresses some of the limitations of zk-SNARK and provides scalable and computational efficiency and secure verification. Its main features, besides zk-SNARK, are that it is transparent and does not require preprocessing. With zk-STARK, complex calculations can be produced at a minimum cost. Since it does not require a secure installation, it eliminates creating secret parameters that can be captured during an attack. This feature makes its use in open blockchain systems more advantageous than zk-SNARK. It is also thought to provide PQ-security [66]. zk-STARK is scalable and uses advanced techniques such as polynomial evaluation and error correction codes to reduce the proof size and increase confirmability. It includes key components such as the Fast Reed-Solomon Interactive Oracle Proofs of

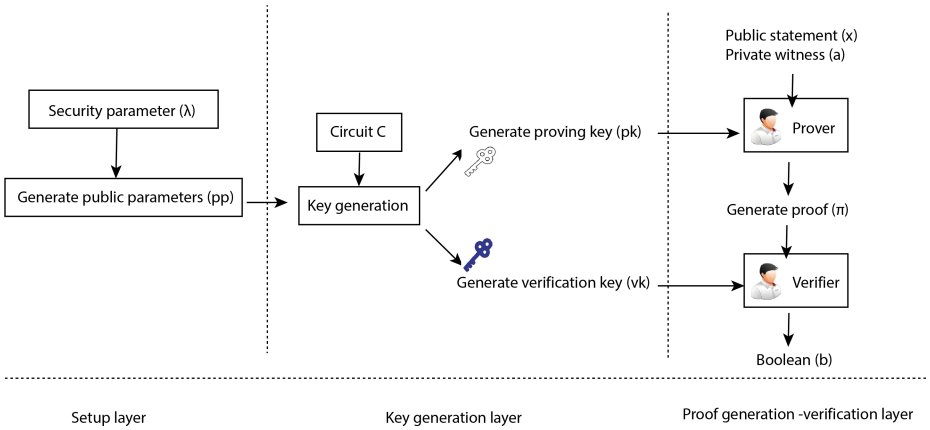

FIGURE 4. The working principle of zk-SNARK.

Proximity (FRI) protocol and low-degree testing (used to verify the correctness of the calculation efficiently), which is an effective protocol for low-degree testing in validating the evidentiary claim.

ZKP is used in real-world systems such as IoT, healthcare, and supply chains to protect data authenticity, prove the authenticity of medical data while preserving its confidentiality, or reduce the risk of data leakage in identity verification [57]. In PQB systems, ZKP provides an important security layer that strengthens user privacy at its core by allowing the user to prove the authenticity of a transaction without revealing their private key when signing a transaction. In other words, it protects data integrity and transaction security. It also helps to provide complete anonymity on the blockchain by ensuring that only necessary information is shared, minimizing users' sensitive data disclosure, and hiding data such as sender-receiver addresses, transaction amounts, and transaction links [67]. Therefore, lattice-based signature algorithms such as post-quantum secure Dilithium can be combined with ZKP to protect the privacy of transaction identities and optimize signature verification processes. It can be strengthened in terms of both security and privacy. Solutions such as ZKP-based zk-rollup minimize the transaction load on the blockchain by producing a single proof for many transactions and provide performance in addition to privacy in PQB systems. In addition, zk-STARK protocols, which use hash-based and polynomial commitments in their structure, can provide high performance and quantum resistance in addition to privacy in PQB systems [68].

However, integrating these techniques into real-world systems brings practical challenges and considerations, such as interoperability between networks and adapting to different consensus mechanisms and cryptographic standards [53]. ZKP's protocols affect the computational cost by imposing overhead, especially in resource-constrained IoT systems. In addition, integration into existing systems can require time and resources to ensure compatibility with these infrastructures, which can cause compatibility challenges. Therefore, integration processes should address these challenges in providing ZKP-based PP-PQB systems.

This section discusses in detail the integration of ZKP with post-quantum digital signatures. Its contributions to data integrity and transaction verification processes in PQB systems explain its privacy and security-enhancing role. In addition, the advantages and limitations of ZKP protocols in terms of scalability, transaction cost, and security are evaluated.

IV. SYSTEMATIC REVIEW REPORT ON STUDIES

This section presents notes on the studies determined after EC on the identified journal papers and the answers to the RQs. The contributions, advantages, and limitations of privacy-preserving techniques proposed for enhancing blockchain reliability and resisting quantum attacks are discussed. Figure 1 illustrates the recent upward trend in research related to PP-PQB, highlighting the novelty of this field. Table 5 briefly summarizes the contributions of privacy-preserving studies in PQB-related studies in journal papers obtained after specific filtering, as mentioned in Table 4, such as conference papers, duplicate studies, and abstract reading. Table 6 discusses the general advantages and disadvantages of the obtained studies. Table 7 shows a summary of the protocols used in the studies. Table 8 summarizes the methods used to increase confidentiality and integrity in the studies. Table 9 provides the effects of the techniques used in the studies on OISS. Table 10 discusses the challenges arising from the methods used in the studies and their impact on real-world usage.

A. REVIEW OF THE SELECTED PAPERS

In [69], the authors proposed the concept of Blockchain Designated Verifier Proof (BDVP) and designed the scheme, which was called the privacy-preserving zero-knowledge proof (ZKP) scheme for blockchain applications. The scheme emulates a false secret to simulate verifier proof. They also proposed a post-quantum solution using the Lyubashevsky

scheme based on the SIS problem. This approach includes innovative elements and methodologies, such as the use of the chameleon hash function and techniques based on the discrete logarithm problem (DLP), which contribute to the novelty of the research.

In [70], the authors aimed to provide unbiased and randomness through threshold verifiable random functions (VRF) for proof-of-stake (PoS) blockchain. They provided a compiler for achieving PQ-VRF from a classical VRF solution using symmetric-key primitives, validated through the quantum-safe zero-knowledge systems ZKBoo and ZKB++. VRF satisfies several critical properties such as consistency, domain range correctness, probability, and uniqueness, making it robust and reliable even in the presence of quantum computing threats. Their proposed random beacon solution via VRF offers a secure concept of randomness for properly integrating consensus protocols and other cryptographic applications.

In [67], the author proposed a new post-quantum secure lattice-based aggregate signature scheme specifically for Bitcoin. The paper provided a scheme based on Dilithium and zk-STARK protocol to minimize the increase in signature size. A lattice-based collective signature scheme was developed using the zk-STARK protocol and the efficient Dilithium implementation. The zk-STARK was used to increase efficiency.

In [71], the authors presented a structure that uses a post-quantum threshold algorithm resistant to quantum attacks on private keys to manage blockchain data. The threshold algorithm is based on NTRU (resistant to quantum attacks due to the complexity of the problems in the lattice) and is resistant to quantum attacks such as Grover's and Shor's. A threshold algorithm was used in the system to reduce the risk of centralization on a key basis. This method ensured that the private key was divided into parts for privacy protection and could be solved by participants. Thus, it enabled the key management encryption to be completed jointly and data authorization to be completed.

In [72], the authors presented an audit mechanism using blockchain that eliminates certificate problems with a lattice-based and identity-based cryptosystem (IBC) in a cloud storage environment. The data audit mechanism based on lattice-based cryptography provided data integrity resistant to quantum attacks. With blockchain, the activities of third-party controllers (TPA) were monitored, and the central audit burden was reduced. An identity-based structure was used for certificate complexity.

In [73], the authors proposed a new traceable lattice-based traceable ring signature (TRS) scheme with the TripleRing structure. In the scheme, the TRS structure was improved and anonymized for resistance against quantum attacks. The anonymous signature system is based on SIS/LWE problems. The TripleRing structure was developed by improving the DualRing structure with short signature sizes, achieving shorter signature sizes and faster signature generation times.

In [74], the authors developed a post-quantum-based privacy-preserving access control method for the Industrial Internet of Things (IIoT). The dynamic access control method uses the ERC721 smart contract non-fungible token (NFT). A post-quantum encryption algorithm called NTRU is used for user privacy, which is resistant to quantum attacks and efficient for resource-limited IIoT devices. Malicious record attacks are prevented by the integrated time interval on the contract.

In [75], the authors presented a blockchain-integrated data management system in logistics systems and designed a lattice-based undeniable signature scheme against quantum attacks. To prevent forgery or false declaration of user identities in the system, a signature-authentication mechanism was developed, and data security was provided by using post-quantum cryptographic problems such as SIS and LWE.

In [76], the authors designed a linkable ring signature (LRS) scheme that provides post-quantum security. The security of the verification was provided with lattice-based problems such as SIS and LWE, and a structure resistant to adversary-chosen-key attacks and tamper-proof was presented to prevent the creation of fake signatures. The study does not include direct blockchain integration, but it was designed as an easily integrated solution for security problems such as manipulating signatures, forgery, and privacy violations in blockchain-based systems.

In [77], the authors proposed a post-quantum secure blind signature scheme against quantum computer attacks in blockchain applications. The scheme used bimodal Gaussian distribution and rejection sampling to protect sensitive information. The security of the signature scheme was proven using the random oracle model. It was proposed for signature verification in blockchain-based systems.

In [78], the authors presented a configuration of the zk-SNARK protocol to provide post-quantum security. In the study, an analysis was presented on how to transform zk-SNARK proofs into Rank-1 Constraint Satisfiability (R1CS) relations to obtain efficient computations with polynomial-based computations using single and multivariate polynomial (Polaris, Spartan) encodings. It was stated that the applied method improved the performance of R1CS encodings. They stated that post-quantum secure zk-SNARK was constructed with security based on symmetric key schemes. Lagrangian interpolation was used for efficient computation of polynomials and the efficiency of the interpolation was increased by selecting the interpolation sets as sub-domains. Although the study does not directly integrate blockchain applications, the proposed polynomial-based zk-SNARK approach has been developed to provide privacy-security in PQB systems.

In [79], the authors proposed a quantum-resistant blockchain-based system that preserves privacy by developing the data deduplication protocol (QBDD). In the study, R-LWE encryption, signcryption, and proxy re-encryption techniques were used to provide data security, verify the

integrity of the message, reduce data redundancy, and create a structure resistant to quantum attacks in the vehicular crowdsensing system. They used a blockchain-based system to increase the efficiency and fairness of the system.

In [80], the authors proposed a blockchain network-based Self-Sovereign Identity system design that provides post-quantum security integrated into the Open Radio Access Networks (O-RAN) architecture. They presented a BCN-based solution for a secure authentication and resource-sharing mechanism between multiple mobile network operators (MNOs). Decentralized Identifiers (DIDs) and verifiable credentials were used for authentication and authorization in SSI. No specific PQC algorithm was used in the system. However, it aimed to optimize PQC calculation methods based on high-degree polynomial multiplication by using the Toom-Cook multiplication method to accelerate mathematical calculations of PQC algorithms.

B. ANSWERS TO THE RQS

1) RQ1 WHAT ARE THE MAIN METHODS FOR PP-PQB?

PQB aims to protect the privacy of users by meeting both privacy and security requirements. Techniques such as blind signature, bimodal Gaussian distribution, rejection sampling, and ZKP that provide anonymity have been developed and implemented to protect of users sensitive information and perform secure transactions [81], [82]. In this context, it is expected that blind signature, ZKP, hybrid cryptography, secure sharing of encryption keys with quantum key distribution (QKD), data encryption and anonymization methods, consensus mechanisms, and smart contract security methods will be developed for PP-PQB systems to provide resistance against quantum attacks. Developing these technical methods is critical to increasing the resilience of blockchain-based systems against quantum attacks while preserving privacy. This section provides an overview of the techniques employed for PP-PQB in the reviewed studies and a summary of the methods (Table 7).

In [69], the authors provided a method to verify the proof of pre-determined validators and protect this process from third parties. It allowed the verifier to verify and simulate the authenticity of proof using the chameleon-hash function. It provided post-quantum security by developing the Lyubashevsky scheme based on the SIS problem and utilizing lattice-based encryption techniques. In the method they adopted, With the proposal of a Non-Interactive Quantum Resistant (NIQR) protocol, they aimed to reduce the number of interactions between the prover and verifier in the ZKP process and increase security and privacy with a new protocol design called BDVP.

In [70], a practical PQ-VRF for the quantum resistive PoS consensus layer was presented. The work leverages ZKBoo and ZKB++ with a quantum-resistant hash-based signature. These advanced encryption protocols used multi-party computation (MPC) techniques (ZKBoo) and NIZK proofs (ZKB++), which significantly reduce their size. The study was combined cryptographic techniques with symmetric key fundamentals to obtain PQ-VRF and integrates VRF into PoS.

In [67], a new lattice-based aggregate signature scheme (LAS) method was proposed by using Dilithium and zk-STARK. This method allowed for minimizing the increase in signature size, which was significantly reduced compared to using Dilimhium alone. With the zk-STARK integration, which is a ZKP structure, user privacy was ensured, and aggregate signature verification was simplified and accelerated. The method they adopted aims to increase post-quantum security and transaction efficiency, in the case of Bitcoin.

In [71], the methods used are NTRU encryption, linear secret sharing scheme (LSSS), and threshold algorithm. The security of the modified NTRU encryption relies on the difficulty of solving lattice-based problems, particularly the shortest vector problem (SVP) in lattice cryptography, which provides resilience against both classical and quantum attacks. Data was encrypted by dividing it among multiple participants, and threshold encryption was used to eliminate the use of centralized management of private keys in decryption and to cooperate with a certain number of participants. LSSS was used to decrypt a certain threshold number of participants, and the private key is shared with a linear secret-sharing matrix.

In [72], the proposed method used lattice-based cryptography, SIS-based techniques, andIBC. These methods were integrated to provide post-quantum security for data integrity checking and secure key management.IBC was used for secure key generation based on user identities. Blockchain was used to audit the activities of TPA and cloud service providers.

In [73], the proposed method introduced a novel lattice-based TRS using a TripleRing structure, which was designed to enhance privacy protection against quantum attacks and implemented in a quantum-resistant blockchain model. TRS was used to prevent the user from signing more than once and to provide anonymity. DualRing was used to make the signature size efficient by reducing it, and TripleRing was used to make the signature more secure by adding traceability. Blockchain was used as an access control mechanism.

In [74], the proposed method employed NTRU encryption, time interval functions (TIF), and smart contracts. NTRU encryption provided post-quantum security to protect user information against quantum attacks, while TIF was used to prevent malicious registration attempts. Smart contracts were utilized to manage various transactions. These methods aim to provide post-quantum secure access to IIoT devices.

In [75], a lattice-based undeniable signature scheme was designed using the NTRU lattice structure based on the SIS problem. The method was supported by techniques such as the bimodal Gaussian distribution and the randomized nearest-plane algorithm, which are considered computationally hard problems.

TABLE 5. Contributions of the selected studies.

Study	New protocol design	Main Contribution
[69]	✓	Leveraging Chameleon hash functions, DLP-based security, and the combination of ZKP with signature schemes to provide a robust scheme called BDVP.
[70]	✓	It provides shorter signatures and signing keys using ZKBoo and ZKB++ quantum-safe zero-knowledge systems for low communication complexity.
[67]	✓	Securely verifying multiple signatures and increasing post-quantum computing efficiency using zk-STARK and aggregate lattice-based signatures.
[71]	✓	It offers an advanced structure to ensure data privacy by providing a threshold structure that provides post-quantum security in blockchain-based systems.
[72]	✓	An audit mechanism structure that provides data integrity-confidentiality using lattice-based structure, identity-based cryptography (IBC), and blockchain-based in cloud storage.
[73]	✓	A post-quantum secure blockchain model has been developed with the proposal of the TripleRing structure, which offers shorter signatures and higher efficiency by improving the TRS structure.
[74]	X	A privacy-preserving post-quantum-based access control system for IIoT based on the ERC721 standard.
[75]	✓	The design of a post-quantum secure blockchain architecture by developing an undeniable signature scheme against quantum attacks for data privacy and security in cold chain logistics systems.
[76]	✓	A post-quantum secure LRS framework that provides unconditional anonymity against quantum attacks using Lattice-based cryptography for signature security.
[77]	✓	The development of a secure lattice-based blind signature scheme against quantum attacks for blockchain-based systems.
[78]	✓	Providing post-quantum security focused on examining zk-SNARKs with univariate/multivariate polynomial codes and transforming them into R1CS relations.
[79]	✓	Development of an R-LWE-based post-quantum secure blockchain protocol integrating signcryption and proxy re-encryption protocols for vehicular crowdsensing systems.
[80]	X	Blockchain-integrated PQC-based SSI system for secure and decentralized identity management in O-RAN.

In [76], the OrthoTrapGen algorithm, the NIWI proof system, the OTS, and the SampleRight algorithm were employed as key components in the post-quantum secure LRS signature scheme. OrthoTrapGen was used to create cryptographic trapdoors, enhancing the security of the signature scheme. The NIWI proof system ensured signer anonymity within the LRS framework, while OTS prevented the reuse of signatures. Finally, the SampleRight algorithm was used to securely generate random secret keys, further securing the signature process.

In [77], an post-quantum blind signature scheme was designed using a lattice-based blind signature, Bimodal Gaussian distribution, rejection sampling, and random oracle model methods. The blind signature was used for message signing by hiding user identity. Gaussian distribution is used to make the blinding process more secure. Rejection sampling was used to prevent rejection and increase signature security.

In [78], methods such as zk-SNARK structure, Uni/Multivariate Polynomial Coding, Fast Reed-Solomon IOPs (FRI), Lagrange Interpolation, and Fiat-Shamir Transform were used to provide post-quantum security against quantum attacks. In Spartan and Polaris schemes in zk-SNARK structures, Polynomial operations, Hash functions, and Fiat-Shamir transformation parts were optimized to provide post-quantum security. Collision-resistant hash (CRH) functions were used to provide post-quantum security. Uni/Multivariate Polynomial Coding was used to increase computational efficiency for Boolean or subdomain circuits by converting R1CS relations into polynomial equations. FRI is used for fast checking in data verification. Lagrange interpolation was used to represent the data array with

TABLE 6. The overall advantages and disadvantages of studies.

	Advantages	Disadvantages
[69]	The BDVP scheme's use of the LBPQ encryption technique and chameleon hash functions ensures both privacy-security and a robust theoretical foundation by protecting sensitive information. The use of NIQR-BDVP contributes to scalability by reducing the number of required interactions.	The use of chameleon hash functions and the Lyubashevsky scheme in the BDVP scheme and ZKP has increased the system's complexity. Additional encryption calculations required in BDVP increase performance overhead and reduce availability by demanding more processing power and resources.
[70]	The use of VRF, ZKBoo, and ZKB++ ensures publicly verifiable randomness and transactions with manageable performance overheads, providing quantum resistance and security. Low communication complexity is achieved by integrating ZKBoo and ZKB++ with PQ-VRF.	The integration of ZKBoo and ZKB++ with PQ-VRF adds extra overhead and increases system complexity. While ZKBoo and ZKB++ reduce communication complexity, the use of PQ-VRF still introduces performance overhead, impacting processing speed and efficiency compared to classical methods.
[67]	zk-STARK integration in the LAS scheme ensures data integrity and confidentiality, while also providing quantum-resistant security. The use of zk-STARK in the LAS scheme increases computational burden and complexity in transaction verification, especially impacting devices with limited resources.	The use of zk-STARK in the LAS scheme increases computational burden and complexity in transaction verification, especially impacting devices with limited resources. Protocols with different encryption standards, particularly those recommended for Bitcoin, may face compatibility issues.
[71]	NTRU ring-based encryption is used to provide resistance against quantum computer attacks. Implementing Linear Secret Sharing Scheme for NTRU keys enables distributed key management (DKM) across multiple nodes, enhancing security.	The overhead from NTRU-based algorithms and threshold encryption negatively impacts transaction time and resource consumption, affecting overall performance. The use of DKM increases system complexity, requiring more nodes and participants for effective management.
[72]	A quantum-resistant, post-quantum secure blockchain mechanism enhances data integrity in cloud storage. The IBC eliminates the need for complex certificate management and enables secure data-sharing control, increasing system performance	The system requires more processing power than classical methods, potentially causing performance issues with large data sets. Dependence on TPA could lead to ongoing reliability concerns, and the lack of full practical testing introduces uncertainty regarding real-world applicability.
[73]	Post-quantum security is provided with SIS and LWE protocols. The TripleRing structure enables shorter signatures and faster generation compared to other post-quantum TRS.	The TripleRing structure and lattice-based signatures increase system integration difficulty and mathematical complexity. Multi-signature processes lead to longer verification and generation times, impacting performance in large-scale blockchain applications.
[74]	NTRU-based encryption provides post-quantum security for secure data transmission in resource-limited IIoT systems. An integrated time interval control function adds an extra security layer.	Complex smart contract transactions lead to increased transaction costs. The need for frequent updates in access control causes network delays and limits scalability.
[75]	A lattice-based undeniable signature protocol on SIS provides resistance to quantum attacks. Data security is enhanced by allowing third-party encryption and transfer using the Proxy Re-Encryption protocol.	SIS-based structures with large key sizes delay transaction times on resource-limited devices, impacting performance. Lattice-based structures and proxy re-encryption require additional infrastructure for key management, increasing costs.
[76]	The LRS scheme offers protection against quantum attacks using lattice-based problems like SIS and LWE. It provides signature anonymity and a strong defense against adversary-chosen key attacks.	Lattice-based algorithms lead to performance and energy issues on resource-limited devices. The LRS scheme's dependency on ring size increases verification time, causing delays in large systems.
[77]	Lattice-based blind signatures provide data security against quantum attacks. Fixed signature size and efficient signing are achieved with Bimodal Gaussian Distribution and Reject Sampling.	Blind signature protocols in this lattice-based system increase verification time, impacting performance during privacy protection. The complexity of the lattice-based structure complicates key management and storage in larger systems.
[78]	Lagrange interpolation and R1CS techniques enable efficient proof production and verification in zk-SNARK for large data sets. The Lagrange interpolation technique reduces production verification costs and enhances security-performance in blockchain integration.	Ensuring security and privacy in post-quantum systems requires high processing power in complex circuits. High transaction traffic leads to performance issues due to large signature sizes and transaction costs.
[79]	R-LWE-based encryption provides post-quantum security against quantum attacks. The Message-Locked Encryption (MLE) protocol controls data duplication between RSUs and the cloud, reducing storage-bandwidth costs and preserving privacy.	Continuous data deduplication in high vehicle traffic scenarios leads to storage issues and increased costs. Proxy re-encryption and signcryption protocols enhance data security but introduce communication delays, impacting performance.
[80]	Post-quantum secure blockchain enhances resource sharing and authentication in O-RAN networks. Blockchain-based SSI protocol ensures privacy and anonymity for personal data.	Lack of smart contract security analysis for trust verification introduces potential vulnerabilities. The Toom-Cook method benefits multi-core processors but impacts performance on single-core processors with limited resources.

polynomials. These polynomials are accelerated with FFT to obtain an efficient computational process. Fiat-Shamir Transform is used to convert interactive proofs into non-interactive ones.

In [79], methods such as grid-based proxy re-encryption, MLE, R-LWE, and RSU were used for the post-quantum blockchain structure against quantum attacks. Grid-based proxy re-encryption was used for data confidentiality integrity and non-repudiation against quantum attacks. It was used to prevent data duplication with MLE.

In [80], the Toom-Cook multiplication method, Hyperledger Indy, DIDComm messaging protocol, and Kubernetes were utilized for O-RAN-integrated, post-quantum secure blockchain. Hyperledger Indy and the DIDComm protocol enabled secure authentication and data transmission within the O-RAN architecture. Kubernetes CPU management policies improved the efficient use of computational resources.

2) RQ2 WHAT METHODS ARE USED TO IMPROVE AUTHENTICATION, CONFIDENTIALITY, AND INTEGRITY IN PP-PQB APPLICATIONS?

Integrating integrity-ensuring protocols like ZKP into PQB applications significantly enhances the accuracy of computations and strengthens the confidentiality, integrity, and authentication of inputs. This situation can be evaluated in three key aspects: enhancing confidentiality (e.g., data minimization), strengthening integrity (e.g., tamper-proof proofs and non-repudiation), and improving authentication (e.g., anonymous authentication). This section discusses the existing methods and components replaced by techniques used for PP-PQB in the reviewed studies, along with a summary of the methods applied to improve authentication, confidentiality, and integrity.

Table 8 lists the methods used to ensure these functions and provides details of protocols developed specifically to withstand quantum attacks, as utilized in current research. Protocols such as ZKP, lattice-based encryption, and NIQR-BDVP are used to strengthen confidentiality and integrity functions. These methods aim to provide security in data verification without disclosing identity information and are particularly focused on post-quantum security. However, integrating these protocols can increase system complexity, potentially impacting energy efficiency and performance.

3) RQ3 WHAT ARE THE EFFECTS OF THE METHODS APPLIED TO PP-PQB APPLICATIONS ON OISS?

In terms of OISS, the impact on the overall architecture of PP-PQB applications has been evaluated, particularly considering the role of ZKP integration within the system. ZKP, by its nature, can reduce overhead on the blockchain by verifying the correctness of transactions without revealing sensitive data. However, proof and verification are computationally intensive, and data related to the proof requires storage, which can increase the computational overhead and storage requirement. This may also affect the computational complexity. Additionally, ZKP supports interaction between networks and has the potential to facilitate interoperability, thanks to the verification of incoming transactions without sharing data. However, the standardization problem of different systems can make integration difficult, and the complexity of ZKP can cause difficulty in integrating into blockchain protocols. If we consider ZKP integration in terms of scalability, it can enable higher volume transactions on the chain as it enables efficient transaction verification. It also has the potential to reduce the overhead on the main chain significantly, thanks to layer-2 solutions called zk-rollup, which can verify multiple transactions with a single proof. However, proof generation time can slow down processing efficiency, and increased latency can affect the processing speed of the overall system.

The evaluation of security, which is the last stage of OISS, with the ZKP integration, ensures that the data is not rejected and increases confidentiality, as its accuracy is verified without being disclosed. However, ZKP is challenging to implement correctly, and integration errors may cause security vulnerabilities. ZKP integration in PQB provides significant benefits for OISS. However, the concept to be proposed will bring additional overhead and complexity and requires careful evaluation and integration to balance these effects. Table 9 shows the effects of the methods used in the studies on OISS. We evaluate the studies we obtained from the EC in terms of OISS;

Overhead: In PQB-based studies, the complexity of cryptographic operations and the increase in data processing times lead to higher transaction fees or reduced efficiency, creating additional resource consumption.

Interoperability: In PQB-based studies, interoperability refers to the seamless operation of different blockchain networks or integrated systems. However, the encryption standards, key management systems, and security protocols used in PQB can make it challenging for different networks to work in harmony with these systems, potentially requiring additional protocols to ensure interoperability.

Scalability: In PQB-based systems, it is proportional to the capacity of the network to effectively meet the increasing demands and transaction volume in terms of managing large transaction volumes and solving performance problems.

Security: Ensuring the post-quantum security of PQB-based systems against quantum attacks requires assessing the security of key management, data integrity, signature verification, and communication protocols used within the network; this implies that new security vulnerabilities may arise.

The methods used to solve these problems in classical crypto-based blockchain systems differ from the methods used to solve these problems in PBQ systems. In classical crypto-based blockchain systems, optimizations are made by minimizing transaction times and memory usage in asymmetric cryptosystems such as RSA and ECDSA. Symmetric algorithms (such as AES) are combined with

TABLE 7. Summary of methods used in the studies.

	PQ-security	Specific hash	Integration blockchain	Privacy-Security
[69]	Lyubashevsky's post-quantum signature scheme (PQSS) (based on the SIS and LWE problems)	Chameleon hash function (Ateniese's ID-based chameleon hashing, Trapdoor key)	PoC	NIQR-BDVP, ZKP (DVP)
[70]	Hash-based signature scheme ((PQ-VRF (based on the symmetric-key primitives to ensure short signatures and efficient computation) (Quantum-Secure Distributed Random Beacon (QS-DRB), verifiable secret sharing (VSS))))	SHA-256	PoS	ZKBoo and ZKB++ (type of ZKP) (to derive ZKP from secure multi-party computation (MPC))
[67]	Lattice-based signature scheme (used Dilithium and signature aggregation (LAS)) based on the LWE and SIS problems))	SHA-256	PoW	zk-STARK (type of ZKP)
[71]	NTRU encryption (based on R-LWE)	-	PoC	Threshold Encryption and LSSS
[72]	Lattice-based cryptography (based on SIS) and post-quantum public auditing	-	PoC	IBC
[73]	TRS (based on SIS and LWE)	-	PoC	TripleRing, SIS and LWE
[74]	NTRU (based on R-LWE)	-	PoC	NTRU and TIF
[75]	NTRU (based on SIS)	Random oracle hash function	PoC	NTRU
[76]	LRS (based on LWE and SIS)	-	-	OTS, SampleRight and NIWI proof system
[77]	Lattice-based blind signature (based on SIS and bimodal gaussian distribution)	SHA-256 and RIPEMD-160	PoC	Random oracle model, reject sampling
[78]	Polaris-Spartan (uni/multivariate polynomial encodings), Fiat-Shamir transform and FRI	Minimum multiplicative complexity (MiMC) and collision resistant hash function (CRH)	-	zk-SNARK, Fiat-Shamir transform and FRI
[79]	Signcryption (based on R-LWE) and proxy re-encryption	-	PoC	R-LWE, MLE
[80]	PQC calculation methods based on high-degree polynomial multiplication (based on general PQC calculations)	SHA-256	Hyperledger Indy (RBFT)	DIDComm messaging protocol

asymmetric algorithms, parallelization is performed, or low-cost lightweight cryptography is used for IoT devices. Aggregate signature methods are used due to the PQ signature size and transaction overhead used in PQB-based systems. In addition, NIWI, OTS, Rejection sampling, quantum compatibility layers, and efficient key management systems such as bonsai tree technology or hybrid approaches are used. Signatures such as blind signature, Multisignature, and Threshold Signature are used to make transactions more secure. Bridges and intermediate layers have been developed for data and asset sharing between blockchain networks. Standardized protocols, such as the ERC-20/ERC-721 token standard, increase compatibility between smart contract platforms. In addition, L2 solutions (such as Rollups and Lightning Network) are used to reduce the overhead on the main layer of the blockchain. In PQB, off-chain solutions are used, just as classical blockchains.

4) RQ4 WHAT ARE THE KEY FUTURE CHALLENGES IN ADOPTING AND IMPLEMENTING PRIVACY-PRESERVING STRUCTURES, INCLUDING ZKP, IN PQB IMPLEMENTATIONS?

This section explores the limitations, future directions, and real-world use cases associated with adopting privacy-preserving structures within PQB applications. Beyond the OISS aspects discussed earlier, major challenges include computational efficiency, robustness, complexity, and usability, as detailed in Table 10.

Computational Efficiency: Computational efficiency is assessed based on resource utilization, proof size, and proof generation-verification times. Many studies identified in

TABLE 8. The methods used to enhance confidentiality and integrity.

	Protocol/Method	Privacy Enhancement	Integrity Enhancement	Authentication Enhancement	Brief Description
[69]	BDVP chameleon hash function, Lyubashevsky's quantum-resistant ZKP (based on SIS-LWE)	ZKP,BDVP	Lyubashevsky's quantum-resistant ZKP, Schnorr's ZKP, chameleon hash function	BDVP	A BDVP protocol, a secure PQ privacy preserving ZKP protocol.
[70]	Hash-based signature sch-eme, PQ-VRF	NIZK, ZKBoo, ZKB++, PQ-VRF	VRF, QS-DRB	VRF	A post-quantum secure signature scheme method.
[67]	LAS scheme, Dilithium, zk-STARK	zk-STARK	Dilithium	-	A post-quantum secure LAS scheme using Dilithium and the zk-STARK.
[71]	LSSS and NTRU	NTRU	Lattice-based signature	NTRU, LSSS, DKM, threshold algorithm	Uses NTRU lattice-based threshold algorithm to prevent quantum attacks.
[72]	Lattice-based cryptograp-hy, IBC	IBC	PQ public audit	IBC	A lattice-based blockchain supported audit mechanism.
[73]	Lattice-based traceable ring signature scheme	TripleRing (based on LWE, SIS)	Ring Signature, SIS	TripleRing	Ring signature, TripleRing structure, and digital signatures usage in PQ secure blockchain structure.
[74]	NTRU (based on R-LWE)	NTRU	-	NTRU,TIF	A smart contract token-based access control architecture with NTRU.
[75]	NTRU-based Undeniable signature scheme	Challenge-response, random gaussian distribution	BCCLS, lattice assumption	NTRU, random hash functions	Unrepudiable signature based on NTRU (based the SIS problem) for confidentiality and integrity.
[76]	LRS scheme, OrthoTrapGen, Basis-Rand, NIWI	Signer-anonymity, NIWI	OTS, adversarially-chosen -keys	NIWI, OTS	A Lattice-based LRS scheme resistant to quantum attacks.
[77]	Lattice-based blind signature, bimodal gaussian distribution	One-more Unforgeability, Random Oracle Model	Rejection Sampling, SIS	Bimodal gaussian distribution, rejection sampling	Quantum attack resistant lattice-based blind signature scheme.
[78]	Polaris-spartan protocol, reed-solomon interactive oracle proofs (FRI), goldwasser Kalai rothblum (GKR), Univariate and Multivariate sumcheck protocol	Polynomial masking, fiat-shamir transform	Polynomial commitment schemes, Schwartz-Zippel Lemma, sum-check protocol, FRI and GKR	FRI, IOPs	Developing privacy-enhancing methods in zk-SNARKs for post-quantum security.
[79]	R-LWE-based proxy re-encryption and sign-cryption	Proxy re-encryption	Signcryption	Signcryption, proxy re-encryption	PQB implements R-LWE-based proxy re-encryption for secure data sharing, signcryption for integrity.
[80]	DID, verifiable credentials	DID, verifiable credentials	DID, BSN	DID, verifiable credentials	A secure PQ decentralized identity management system in O-RAN.

Table 10 highlight that ZKP and lattice-based encryption, despite providing privacy, introduce significant resource consumption and computational overhead. For instance, [69] shows that integrating ZKP with chameleon hash functions enhances security but at a cost to computational efficiency, while [70] achieves efficiency improvements using ZKB++ to optimize proof sizes and processing times. These examples suggest a continued need for balancing computational efficiency with privacy requirements, particularly in large-scale PQB systems.

Complexity, Usability, and Real-World Usage: Complexity often arises from additional verification and encryption stages, as seen in various studies within Table 10. In [67], for instance, the zk-STARK integration adds complexity due to enhanced storage requirements, while in [71], NTRU-based threshold encryption increases complexity yet ensures secure key distribution. Real-world usability often depends on designing user-friendly interfaces, especially for resource-constrained environments. Practical applications, such as in IoT or IIoT systems [74], require careful integration of

TABLE 9. The effects of the methods used in the studies on OISS.

	Overhead	Interoperability	Scalability	Security
[69]	X	✓	X	✓
[70]	X	✓	✓	✓
[67]	X	X	✓	✓
[71]	✓	✓	✓	✓
[72]	X	X	X	✓
[73]	✓	✓	✓	✓
[74]	X	X	X	✓
[75]	✓	✓	✓	✓
[76]	X	X	X	✓
[77]	✓	✓	✓	✓
[78]	X	X	X	X
[79]	X	✓	✓	✓
[80]	X	✓	X	✓

complex cryptographic techniques to maintain scalability and usability. However, these methods often raise gas fees and overhead, impacting the usability in high-transaction environments.

Numerous studies in Table 10 show that optimizing computational resources is critical. Methods like zk-rollup for proof efficiency [70] and multivariable polynomial optimizations [78] offer promising directions, but achieving a balance between performance and security remains challenging. Future PQB systems should focus on simplifying complex cryptographic protocols without compromising security. Ensuring compatibility across different platforms [80] and developing user-friendly cryptographic tools are essential to facilitate practical adoption. Applications like smart transportation systems [79] and spectrum sharing in next-generation wireless networks [80] illustrate the growing demand for scalable, privacy-preserving solutions. However, challenges remain in achieving efficient processing times and managing large transaction volumes in real-world PQB implementations.

These challenges underscore the need for ongoing research to optimize computational efficiency, manage complexity, and enhance usability in PQB systems.

V. DISCUSSION, THREATS, AND FUTURE RESEARCH DIRECTIONS

The rapid development of quantum computers and their accompanying communication and data management systems also pose security issues and a serious threat to existing systems [83]. Although studies have been conducted that have the potential to provide solutions resistant to quantum attacks for PQB systems, significant challenges remain in the PQ era. Some current problems that arise in the process of adopting privacy-preserving structures include system complexity, performance and energy efficiency, compatibility issues, and secure data management. Although existing solutions try to alleviate these problems, problems such as the smooth interaction of blockchain networks in the PQ era, the mathematical complexity of PQ signature schemes, and the performance-security balance are still significant challenges, and further research is needed in these areas [26]. The mathematical complexity of PQ-based signature schemes and the computational costs they bring, especially the transaction volume and verification times, negatively affect the performance of blockchain systems. PQB-based systems can cause performance problems, especially in large-scale use of blockchain networks. There are also issues with integrating PQB systems with blockchain networks, especially cross-chain compatibility between traditional blockchain solutions and post-quantum robust structures. Moreover, the scalability problem in blockchain becomes even more complicated with the integration of PQC.

In Table 11, while some components have been made PQ-secure, critical parts such as the consensus mechanism and ZKP still work with classical methods. In addition, there are no hybrid systems where metrics such as performance and security are considered together. These two metrics are mostly optimized separately, but structures such as the security-performance relationship metric model should be addressed in response to the security-performance trade-off challenge for PQB-based systems. In addition, the development of post-quantum secure ZKP schemes and their integration into PQB systems is another important metric that needs to be addressed.

Studies show that integrating post-quantum cryptography into blockchain architectures and ZKP and LBC methods into PQB systems presents various challenges. Lattice-based cryptosystems such as Kyber and Dilithium have larger key sizes than traditional public-key algorithms, leading to more processing time and memory overhead. In addition, key and signature sizes cause delays in data transfer on the network. This can be a critical problem, especially in resource-constrained IoT-based blockchain systems. It also increases the need for data storage in the blockchain and can lead to rapid growth in the total size of the chain. Moreover, implementing PQC in smart contract mechanisms and security protocols can cause delays due to long computational times in applications that require high transaction capacity and result in increased processing power and energy consumption, especially in transaction verification processes. Complex operations in PQC can increase gas costs and make it difficult for users to adopt these systems. Making smart contract languages such as Solidity compatible with PQC may require the development of new compilers or tools. High computational costs can also affect the efficiency of consensus protocols in their working mechanisms. At this point, solutions such as ZKP-based zk-rollup, which improves privacy, can reduce the storage load on the blockchain by compressing transactions. However, ZKP-integrated schemes may increase transaction times when additional security components are used, creating a high computational overhead, and the lack of complete security analysis may pose a risk to system security. Although the ZKBoo and ZKB++ structures, used to provide short signature keys and randomness reliability, are promising, the complexity of these techniques increases computational overhead and can pose challenges in terms of applicability. In addition, NTRU-based and threshold encryption techniques may cause performance

TABLE 10. The impact of real-world use cases and challenges arising from adopting privacy-preserving methods in PQB applications on the studies.

	Computational efficiency	Complexity	Robustness	Usability	ZKP usage	ZKP type	Use case
[69]	X	X	✓	X	✓	Schnorr's ZKP scheme	Identity Verification
[70]	✓	X	✓	X	✓	ZKBoo, ZKB++,	Consensus protocols
[67]	X	X	✓	X	✓	zk-STARK	Banking and finance
[71]	✓	✓	✓	✓	X	-	E-voting, data management, and authorized access
[72]	✓	X	✓	✓	X	-	Data management and integrity control in cloud-based systems
[73]	✓	✓	✓	X	X	-	IoT, Electronic voting systems, E-commerce, and Banking
[74]	X	X	✓	X	X	-	Electric Vehicle Charging Stations and IIoT Based Industrial Applications
[75]	✓	✓	✓	✓	X	-	Cold-Chain Logistics System Digital identity verification and anonymous e-voting systems
[76]	X	X	✓	X	X	-	Digital identity verification and anonymous e-voting systems
[77]	✓	✓	✓	X	X	-	E-voting, IoT security, smart manufacturing, supply chain, healthcare
[78]	✓	X	✓	X	✓	zk-SNARK	Blockchain-based authentication and anonymity systems
[79]	✓	X	✓	X	X	-	Autonomous vehicle networks, smart transportation systems
[80]	✓	X	✓	✓	X	-	Spectrum sharing in O-RAN wireless network systems

TABLE 11. PQB components and evaluations.

Components	PQ	Performance	Usability	Utility
Signature Scheme	Yes	Middle	High	High
Key Exchange	Yes	High	High	High
Consensus Mechanism	No	Low	Middle	Middle
Hash Functions	Yes	Middle	High	High
Traceability Mechanism	Partially	Middle	Middle	Middle
ZKP	Partially	Low	Middle	High

issues in large-scale structures and introduce vulnerabilities against denial of service (DoS) attacks. Finally, lattice-based blockchain integrations face cost and transaction overhead challenges when confronted with high transaction volumes and expanding network structures, which results in additional costs and security risks in auditing processes.

In cryptographic methods, it is important to choose the most suitable algorithm based on the needs of the system and performance metrics such as computational cost, processing times, and resource usage. Computational cost is proportional to the power and time the selected algorithm consumes during processing. This metric is especially critical in resource-constrained IoT devices. Processing time is an important metric in applications that require high-speed processing, such as real-time financial transactions. Another metric, resource usage, is critical in environments such as systems with large data processing capacity or

TABLE 12. Performance metrics of the algorithms in the proposed decision-making structure.

Algorithm	Use case	Computational cost	Processing time	Resource usage	Advantages	Disadvantages
Kyber	IoT, Healthcare	Middle	Middle	Middle	Quantum resistance, IoT compatibility	Large key sizes
Dilithium	Finance, Healthcare	High	High	High	Strong security, efficient signature verification	High computational requirements
zk-SNARK	Finance, IoT	Low	Middle	Middle	High privacy, anonymous payment systems	Performance limitations with large datasets
zk-STARK	Large-Scale Systems	High	Low	Middle-High	Quantum resistance, scalability	High initial setup cost

* Low: Minimal resource requirements. Middle: Moderate resource requirements. High: Significant resource requirements. Very Low: Negligible resource requirements.

resource-constrained IoT devices. Since lattice-based algorithms require large processing power, they must be carefully integrated for real-time applications due to the determined metrics. In addition, large key and signature sizes can create additional overhead in terms of both data transfer and storage. Post-quantum secure lattice-based cryptosystems such as Kyber or Dilithium, which are integrated into protocols optimized for such systems, are suitable. In addition, in ZKP protocols that provide privacy, zk-SNARK may be a disadvantage for large-scale systems that perform more transactions than the zk-STARK structure. Slower, more secure algorithms may be preferred in areas with low transaction frequency, such as health records. The proposed decision-making structure is detailed in Table 12 by classifying the algorithms according to their performance metrics. However, NIST-approved post-quantum algorithms such as Kyber, Dilithium, and FALCON are not yet optimized for integration with blockchain frameworks like Ethereum or Hyperledger. Ethereum has an EVM structure that supports traditional cryptographic protocols for executing transactions. This integration can significantly increase the transaction time and data storage requirements on Ethereum. Especially for smart contracts, using these algorithms can increase the existing gas fees and negatively affect the user experience. Hyperledger Fabric's modular architecture facilitates the integration of different algorithms. However, in large-scale networks, the integration of these algorithms can create an additional overhead on transaction throughput and network performance. Therefore, challenges such as the current determined transaction time or storage impact continue to be the current challenges in this integration. Furthermore, these algorithms may be vulnerable to side-channel attacks, especially in hardware-based implementations [84]. Such attacks aim to access sensitive information through side channels such as power consumption, electromagnetic leakage, or timing information. Moreover, in real-world applications, there is a risk of being vulnerable to attacks due to coding errors, verification processes, vulnerabilities that may occur during key management, and non-optimized application processes [85].

Considering the existing studies examined, they usually focus on only certain components of PQB systems, such as digital signatures or key exchange. These studies cannot provide fully integrated solutions covering all components. This leads to the lack of fully post-quantum secure blockchain architectures. Moreover, it has been observed that hybrid systems that combine traditional blockchain solutions with post-quantum systems to ensure compatibility have not been sufficiently investigated. It has also been observed that the large key sizes and high computational requirements of lattice-based and other post-quantum cryptographic algorithms create scalability problems, and existing studies do not sufficiently address how to manage this balance. ZKP plays a critical role in privacy in PQB systems. However, the development of quantum-resistant ZKP protocols is still in the early stages. Current implementations create a computational burden that is too high to be used in large-scale systems. It has been observed that most studies are theoretical and do not offer a limited number of practical tests or real-world applications. It has also been observed that consensus protocols have not been sufficiently addressed with quantum-resistant innovations.

Considering all these situations, it has been concluded that the solutions applied in the literature to provide this performance-security balance can be improved. Based on this, future research directions and open problems in the existing literature regarding the integration of PP-PQB applications have been identified.

1) Using the standardized protocols and developing communication protocols that provide interoperability across chains are important issues in PP-PQB studies as a future research guideline.
2) Investigating optimization techniques to reduce computational overhead without compromising security

in architectures built using PQC techniques and privacy-preserving structures (such as ZKP).

3) Production verification of integrated ZKP types can also optimize the layer-based architecture for computational efficiency to reduce the overhead.
4) The use of ZKP in post-quantum symmetric and code-based structures is limited in the current literature and has potential research opportunities for PP-PQB.
5) Developing user-friendly cryptographic tools and interfaces for usability in practical applications built using PQC techniques and privacy-preserving structures can contribute to adopting more effective integration in blockchain applications.
6) Systems based on a single chain have limitations in terms of scalability and efficiency. Therefore, integrating multi-chain models can contribute to efficiency and scalability by enabling components in the system to operate on different chains.
7) Post-quantum-resistant versions of bridges used in post-quantum blockchains must be developed, and hybrid protocols must be developed to integrate with off-chain systems.
8) Layer2 solutions such as Lightning Network, which are used for efficiency and performance in classical blockchain-based solutions, can be made quantum-resistant in PQB systems. In cases requiring high transaction volume, the security and performance of the main chain can be maintained.
9) Integrating algorithms such as Kyber and Dilithium in resource-constrained IoT environments can improve Number Theoretic Transform (NTT) performance by using reduced processing cycles and low-precision computational methods to reduce the computational overhead.
10) To implement the lattice-based cryptographic algorithms recommended by NIST securely, hardware and software solutions resistant to side-channel attacks should be developed, and formal verification techniques should be applied on an application basis.
11) In cases where more substantial key sizes are required to integrate lattice-based algorithms in IoT environments, fully secure versions can be implemented on central servers with a multi-layered structure that allows the algorithm to choose different security levels according to resource availability.

Future research should also address these issues, such as ensuring cross-chain compatibility, increasing scalability without compromising security, and developing new methods for designing and compatibility with user-friendly systems and post-quantum security. Further studies on how applications perform in real-world scenarios are also important.

This section discusses the challenges faced by current studies and future research directions. In particular, suggestions are provided for optimizing post-quantum algorithms and improving the performance of ZKP in resource-constrained IoT devices.

VI. CONCLUSION

Traditional blockchain protocols, which are vulnerable to quantum attacks, can significantly increase security and trust with post-quantum cryptography. PQC techniques and privacy-preserving structures in PQB applications, which have attracted recent attention, have the potential to strengthen privacy and security. Therefore, research on privacy-preserving structures has become the new goal in PQB applications. In our study, we present a systematic review with RQ regarding using privacy-preserving structures in PQB applications. We specifically focus on ZKP applications that provide enhancing features for PQB. In the systematic report, we explain the main methods for PP-PQB (RQ1), the impact of privacy-preserving structures on confidentiality and integrity in PQB applications (RQ2), the impact of the solutions on OISS (RQ3) and how they are evaluated, and the future steps of using privacy-preserving structures in PQB applications (RQ4). Our findings indicate that research studies for using privacy-preserving structures in PQB applications are new, and research addressing the current issue has deficiencies in OISS solutions of PP-PQB. We hope our study can help further research-adoption of using privacy-preserving structures in PQB applications. In future studies, we plan to implement layer-based architecture to solve OISS challenges and improve privacy when using privacy-preserving structures in PQB applications.

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